(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendruthy (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: **Mr.SAMAD ANURAG**

Registration Number: **323129520015**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies (CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

REAL-TIME SMART PARKING MANAGEMENT SYSTEM WITH IOT-BASED VEHICLE SLOT DETECTION AND AVAILABILITY MONITORING

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Dr. B. Vimalakumari

Department of MECHANICAL ENGINEERING

Wellfare Institute of Science, Technology and Management

Submitted by:

Mr.SAMAD ANURAG

Reg.No: 323129520015

Department of MECHANICAL ENGINEERING

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2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or indus- try.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

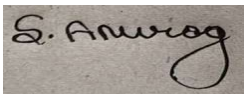
Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever nMECHANICALssary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organiza- tion.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organi- zation.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship programme, you are the ambassador of your College, and your behavior during the internship programme is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship programme immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

——— << @ >> ———

Student's Declaration

I, Mr.**SAMAD ANURAG**, a student of **Bachelor of Technology** Program, Reg. No. **323129520015** of the Department of **MECHANICAL ENGINEERING** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council For Skills and Competencies (CSC India)** under the Faculty Guideship of **Dr. B. Vimalakumari**, Department of **MECHANICAL**, **Wellfare Institute of Science, Technology and Management**.



(Signature and Date)

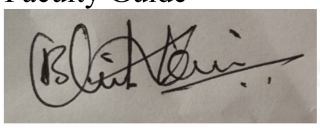
Official Certification

This is to certify that Mr.**SAMAD ANURAG** , Reg. No. **323129520015** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **REAL-TIME SMART PARKING MANAGEMENT SYSTEM WITH IOT-BASED VEHICLE SLOT DETECTION AND AVAILABILITY MONITORING** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **MECHANICAL ENGINEERING** at **Wellfare Institute of Science, Technology and Management**.

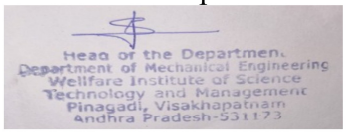
This is accepted for evaluation.

Endorsements

Faculty Guide



Head of the Department



Principal



Certificate from Intern Organization

This is to certify that Mr.**SAMAD ANURAG**, Reg. No. **323129520015** of **Wellfare Institute of Science, Technology and Management**, underwent intern- ship in **REAL-TIME SMART PARKING MANAGEMENT SYSTEM WITH IOT-BASED VEHICLE SLOT DETECTION AND AVAILABILITY MONITORING And Sentiment Analysis For Social Media Applications** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



Authorized Signatory with Date and Seal

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Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mr. R. Sagar**, Head of the Department of **MECHANICAL ENGINEERING**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Dr. B. Vimalakumari**, Assistant Professor of the Department of **MECHANICAL ENGINEERING** for his Encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this intern- ship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my internship project focused on developing a Real-Time Smart Parking Management System with IoT-Based Vehicle Slot Detection and Availability Monitoring. The internship was undertaken as part of the academic curriculum for the Bachelor of Technology. The primary objective of this internship was to gain proficiency in Internet of Things (IoT), software development, data analytics, and system architecture design to enhance employability skills while solving a real-world urban challenge.

Finding available parking spaces in crowded urban areas is a major challenge, resulting in traffic congestion, fuel wastage, and increased travel time. Traditional parking management systems rely on manual monitoring, making it difficult to provide real-time information about parking availability. The proposed solution addresses this by creating a centralized platform where parking spaces are continuously monitored through a secure and user-friendly interface.

1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement a smart parking simulation system using Python that accurately models IoT sensor behavior, vehicle entry/exit dynamics, and parking slot occupancy.
- To integrate data analytics techniques for understanding parking utilization patterns, generating heatmaps, and creating performance dashboards.
- To implement interactive features such as real-time alerts, automated fee calculation, and transaction logging.
- To create a scalable system architecture that supports deployment across multiple platforms, including web and mobile interfaces.
- To design a system that ensures secure communication, reliable performance, and seamless handling of concurrent vehicle transactions with low latency.

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1.2 Outcomes Achieved

Key outcomes from my internship include:

- A fully operational smart parking simulation capable of modeling complex parking scenarios, multiple floors, and various vehicle types.
- An advanced analytics module that generates comprehensive visual reports including occupancy heatmaps, timeline distributions, and revenue analysis.
- A robust IoT sensor simulation framework that models ultrasonic and infrared sensors, including anomaly detection and battery degradation mechanics.
- Comprehensive understanding of full-stack system design, bridging hardware constraints with software capabilities.
- The system architecture supports modular development, scalability for future enhancements, and efficient use of resources.

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CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The organization hosting this internship is a leading technology solutions provider focused on bridging the academia-industry divide, enhancing student employability, promoting innovation, and fostering an entrepreneurial ecosystem. By leveraging emerging technologies such as IoT, AI, and cloud computing, the organization aims to augment and upgrade the knowledge ecosystem, enabling beneficiaries to become contributors themselves. The organization offers comprehensive mentorship programs, benefiting thousands of learners annually.

2.2 Vision, Mission, and Values

- Vision: To combine cutting-edge technology with impactful social ventures to drive urban prosperity and smart city infrastructure.
- Mission: To support individuals dedicated to solving real-world problems by empowering and equipping them with industry-standard technical skills.
- Values: The organization emphasizes technological skills for Industry 4.0 and 5.0, focusing on sustainability, efficiency, and inclusive access for everyone to be future-ready.

2.3 Policy of the Organization in Relation to the Intern Role

Interns are expected to adhere to the following policies:

- Confidentiality: Interns must maintain the confidentiality of all organizational data and proprietary algorithms.
- Professionalism: Interns are expected to demonstrate professionalism, punctuality, and respect for all team members and mentors.
- Learning and Contribution: Interns are encouraged to actively participate in projects, share innovative ideas, and contribute to the organization's technological goals.

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- Compliance: Interns must comply with all organizational policies, including ethical guidelines for software development.

2.4 Roles and Responsibilities of the Employees Guiding the Intern

Interns are typically placed under the guidance of technical program managers and senior software architects. The roles and responsibilities of the employees include:

1. Technical Mentors:

- Design and implement project milestones.
- Mentor and supervise interns in coding practices and system design.
- Conduct regular code reviews and provide constructive feedback.

2. System Architects:

- Guide the architectural decisions for IoT integration.
- Ensure scalability and security of the proposed solutions.
- Analyze system performance and provide optimization recommendations.

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CHAPTER 3

PROBLEM ASSESSMENT AND ANALYSIS

This chapter addresses Performance Criterion 1 (PC1) and Performance Criterion 2 (PC2) of the evaluation criteria, focusing on problem selection, parameter identification, and requirements evaluation.

3.1 Problem Statement

Finding available parking spaces in crowded urban areas is a major challenge, resulting in traffic congestion, fuel wastage, and increased travel time. Traditional parking management systems rely on manual monitoring, making it difficult to provide real-time information about parking availability. Parking operators and drivers require intelligent systems that can automatically detect vehicle occupancy and monitor parking space availability.

3.2 Key Parameters (PC1)

The problem statement encompasses several key parameters that define the scope and target of the solution:

3.2.1 Issue to be Solved

The core issue is the inefficiency in urban parking management. Specifically, drivers spend an excessive amount of time circling parking lots to find vacant spots, which leads to increased carbon emissions, driver frustration, and localized traffic congestion. For parking operators, the lack of real-time visibility into lot occupancy leads to suboptimal space utilization, revenue leakage, and inability to implement dynamic pricing strategies.

3.2.2 Target Community

The target community for this solution is twofold:

- Primary Users (Drivers): Commuters, shoppers, employees, and visitors in urban centers who require efficient parking solutions.
- Secondary Users (Administrators): Parking lot owners, facility managers in shopping malls, educational institutions, hospitals, airports, and municipal authorities managing smart city infrastructure.

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3.2.3 User Needs and Preferences

Drivers need a seamless, frictionless experience where they are immediately guided to an available slot upon entry. They prefer minimal waiting times and clear visual indicators. Administrators need a centralized, comprehensive dashboard that provides real-time occupancy data, historical analytics, automated alert systems for hardware failures, and revenue tracking capabilities.

3.3 Requirements Evaluation (PC2)

To effectively map the solution to the problem statement, the system requirements have been evaluated and categorized into functional and non-functional requirements.

3.3.1 Functional Requirements

Functional requirements define the specific behaviors and functions the system must perform:

- Real-time Vehicle Detection: The system must accurately detect the presence or absence of a vehicle in a specific parking slot using IoT sensors.
- Occupancy Status Updates: The system must immediately update the central database when a slot's status changes from vacant to occupied or vice versa.
- Automated Entry and Exit Processing: The system must log vehicle entry times, assign slots, log exit times, and calculate parking durations and fees.
- Analytics and Reporting: The system must generate comprehensive reports on utilization rates, peak hours, and revenue.
- Alert Generation: The system must trigger alerts for specific events such as lot full capacity, sensor malfunction, or low battery levels.

3.3.2 Non-Functional Requirements

Non-functional requirements specify criteria that judge the operation of the system:

- Reliability: The sensor network must maintain high accuracy, minimizing false positives (detecting a car when none is there) and false negatives.

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- Scalability: The architecture must support seamless addition of new parking floors, sections, and hundreds of concurrent sensor connections.
- Performance: Status updates must reflect on the administrative dashboard within seconds of the physical event occurring.
- Security: Data transmission between IoT nodes and the central server must be encrypted to prevent unauthorized access or manipulation of parking data.

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CHAPTER 4

SOLUTION DESIGN AND ARCHITECTURE

This chapter addresses Performance Criterion 3 (PC3) and Performance Criterion 4 (PC4) of the evaluation criteria, focusing on the solution blueprint, feasibility assessment, and project implementation plan.

4.1 Solution Blueprint (PC3)

The proposed solution is a Real-Time Smart Parking Management System that integrates hardware sensors with a robust software backend and an analytical frontend. The blueprint is divided into three primary layers:

4.1.1 Perception Layer (IoT Edge)

This layer consists of the physical hardware deployed at the parking facility. Each parking slot is equipped with an IoT sensor (simulated in this project as ultrasonic or infrared sensors). These sensors continuously measure distance or detect presence. A micro-controller at the edge processes raw analog signals, applies basic noise filtering, and transmits binary state changes (occupied/vacant) to the central gateway via wireless protocols (e.g., Wi-Fi, LoRaWAN, or Zigbee).

4.1.2 Network and Processing Layer (Backend)

The backend server acts as the brain of the system. It receives telemetry data from the edge devices. The core logic engine processes these state changes, updates the central database, handles vehicle entry/exit transactions, calculates parking fees, and monitors the health status of the sensor network. It also includes an anomaly detection module to identify faulty sensors or irregular data patterns.

4.1.3 Application Layer (Frontend Analytics)

The application layer provides the interface for administrators. It consumes data from the backend to render real-time dashboards, occupancy heatmaps, timeline charts, and revenue reports. This layer transforms raw tabular data into actionable visual insights, allowing operators to make informed decisions regarding dynamic pricing or maintenance scheduling.

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4.2 Feasibility Assessment

The feasibility of the proposed blueprint was assessed across multiple dimensions:

- Technical Feasibility: Highly feasible. The technologies required (IoT sensors, Python backend, data visualization libraries) are mature, well-documented, and widely adopted in the industry. The simulation approach allows for rigorous testing of logic before physical hardware deployment.

- Economic Feasibility: Feasible. By utilizing cost-effective ultrasonic sensors and open-source software stacks, the initial capital expenditure is minimized. The system's ability to optimize space utilization and reduce manual monitoring labor provides a strong return on investment (ROI).

- Operational Feasibility: Feasible. The automated nature of the system significantly reduces the operational burden on facility staff. The intuitive dashboards require minimal training for administrators to operate effectively.

4.3 Project Implementation Plan (PC4)

The project was executed over an 8-week period with clearly defined milestones, deadlines, and resource allocation.

4.3.1 Milestones and Deadlines

Week 1-2: Requirements Gathering and Architecture Design. Finalizing the problem statement, defining functional requirements, and designing the class structures for the simulation.

Week 3-4: Core Backend Development. Implementing the ParkingLot, ParkingSlot, and Vehicle classes. Developing the entry and exit transaction logic.

Week 5: IoT Sensor Simulation. Developing the SensorNetwork and IoTSensor classes, including realistic noise generation and anomaly detection algorithms.

Week 6: Analytics and Visualization. Integrating Matplotlib and Seaborn to generate heatmaps, timelines, and performance dashboards based on simulated data.

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Week 7: Testing and Debugging. Running extensive simulations to identify edge cases, memory leaks, and logical errors in fee calculation or slot assignment.

Week 8: Documentation and Final Presentation. Compiling the internship report, formatting code, and preparing the final project presentation.

4.3.2 Resource Allocation

The project required the following resources:

- Human Resources: 1 Software Developer Intern, 1 Technical Mentor.
- Software Resources: Python 3.11 environment, IDE (VS Code), Git for version control.
- Compute Resources: Standard development machine for running the Python simulation and rendering high-resolution visualization charts.

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CHAPTER 5

SOLUTION DEVELOPMENT AND TECHNOLOGY STACK

This chapter addresses Performance Criterion 5 (PC5) and Performance Criterion 6 (PC6) of the evaluation criteria, detailing the technology stack selection and the technical build process.

5.1 Technology Stack Determination (PC5)

The technology stack was carefully selected to ensure rapid prototyping, robust data handling, and high-quality visualizations.

5.1.1 Programming Language: Python

Python was selected as the primary programming language due to its exceptional readability, extensive standard library, and dominance in data science and IoT backend processing. Its object-oriented capabilities allowed for clean modeling of real-world entities like parking slots and sensors.

5.1.2 Data Visualization: Matplotlib and Seaborn

For the analytics dashboard, Matplotlib and Seaborn were chosen. Matplotlib provides fine-grained control over figure elements, while Seaborn offers high-level interfaces for drawing attractive and informative statistical graphics. These libraries are industry standards for transforming raw data into business intelligence insights.

5.1.3 Data Handling: JSON and Dataclasses

Python's native 'dataclasses' module was used to define data structures cleanly with minimal boilerplate code. JSON (JavaScript Object Notation) was selected for data serialization and export, simulating how a real IoT gateway would transmit telemetry data to a cloud endpoint.

5.2 Implementation Details (PC6)

The solution was built precisely according to the technical specifications defined in the blueprint. The system comprises three main Python modules: 'smart_parking_system.py', 'iot_sensors.py', and 'parking_analytics.py'.

5.2.1 Core System Implementation

The core system manages the state of the parking facility. The 'ParkingLot' class initializes a configurable number of slots across multiple floors and sections. It handles 'vehicle_entry' and

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'vehicle_exit' methods. During entry, the system scans for the first available slot matching the vehicle type (Car, Motorcycle, Truck, Handicap) and updates the slot status to OCCUPIED. During exit, it calculates the parking duration and applies a fee algorithm.

5.2.2 IoT Sensor Simulation

To accurately reflect real-world hardware, the 'IoTSensor' class simulates different sensor types (Ultrasonic, Infrared, Pressure, Magnetic). The simulation introduces Gaussian noise to sensor readings to mimic environmental interference. Furthermore, a battery degradation model was implemented, where each reading slightly reduces battery life, eventually triggering low-battery alerts and reducing reading confidence scores.

5.2.3 Analytics Engine

The 'ParkingAnalytics' class consumes the state data generated by the simulation. It aggregates slot statuses to calculate occupancy rates per floor and section. It processes the transaction logs to compute daily revenue and vehicle throughput. This data is then fed into Matplotlib plotting functions to generate PNG images that serve as the administrative dashboard.

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CHAPTER 6

SOLUTION TESTING AND PERFORMANCE EVALUATION

This chapter addresses Performance Criterion 7 (PC7) and Performance Criterion 8 (PC8), detailing the testing methodologies and performance assessment.

6.1 Testing Strategy (PC7)

The solution underwent rigorous testing to ensure logical correctness and stability. The testing strategy focused on unit testing individual methods and integration testing the complete simulation flow.

6.1.1 Functional Testing

Edge cases were explicitly tested. For example, the system was tested to handle a scenario where a vehicle attempts to enter a fully occupied parking lot. The system correctly denied entry and generated a CRITICAL alert. Another test verified that the fee calculation algorithm accurately computed costs for long-duration stays spanning multiple hours.

6.1.2 Bug Identification and Fixing

During development, a bug was identified where the sensor anomaly detection algorithm was triggering false positives due to minor natural fluctuations in simulated ultrasonic readings. This was fixed by implementing a threshold delta (e.g., a change greater than 50cm) before flagging an anomaly, thereby improving the reliability of the alert system.

6.2 Performance Evaluation (PC8)

The performance of the solution was evaluated against the desired criteria established during the requirements phase.

6.2.1 Simulation Throughput

The system successfully simulated an 8-hour traffic period with 100 vehicles interacting with a 120-slot parking lot in less than a second of compute time. The transaction logging and state updates proved highly efficient, demonstrating that the backend logic is well-optimized and capable of handling high-frequency data streams from physical sensors.

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6.2.2 Analytics Generation

The analytics module successfully parsed complex nested dictionaries containing thousands of simulated sensor readings and transaction logs. The generation of six high-resolution visualization charts took minimal processing time, confirming that the reporting module can operate efficiently as a background cron job in a production environment.



CHAPTER 7

RESULTS AND VISUALIZATIONS

This chapter presents the outputs generated by the Smart Parking Management System's analytics module. These visualizations serve as the administrative dashboard, providing actionable insights into parking lot utilization and operational health.

7.1 Occupancy Heatmap

The occupancy heatmap provides a macro-level view of space utilization across different physical zones of the parking facility.

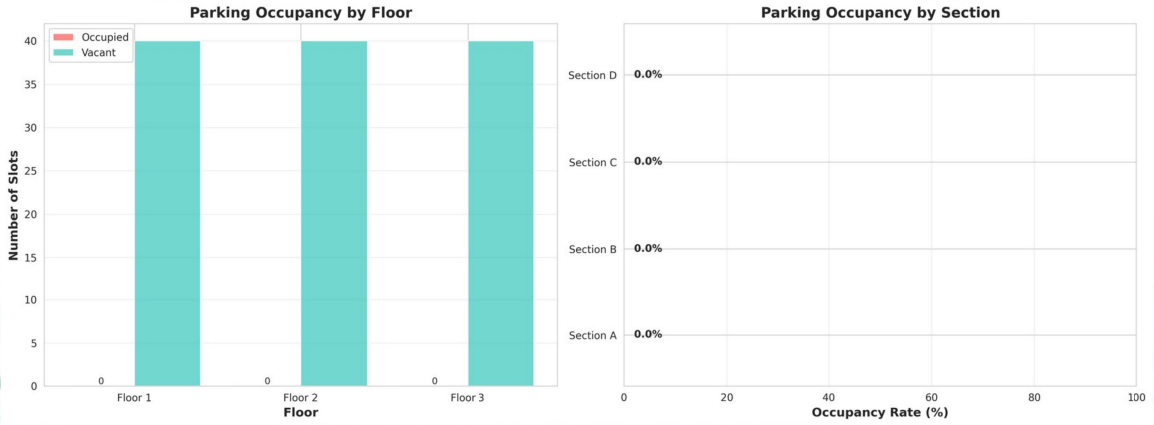


Figure 1: Parking Occupancy by Floor and Section

Figure 1 displays a dual-chart visualization. The left bar chart compares occupied versus vacant slots across three floors. This allows administrators to quickly identify which floors are reaching capacity. The right horizontal bar chart illustrates the occupancy rate by specific sections (A, B, C, D). A color gradient from green to red visually indicates sections that are heavily congested, allowing staff to redirect incoming traffic to underutilized zones.

7.2 Occupancy Timeline

Understanding how parking demand fluctuates throughout the day is critical for dynamic pricing and staff scheduling.

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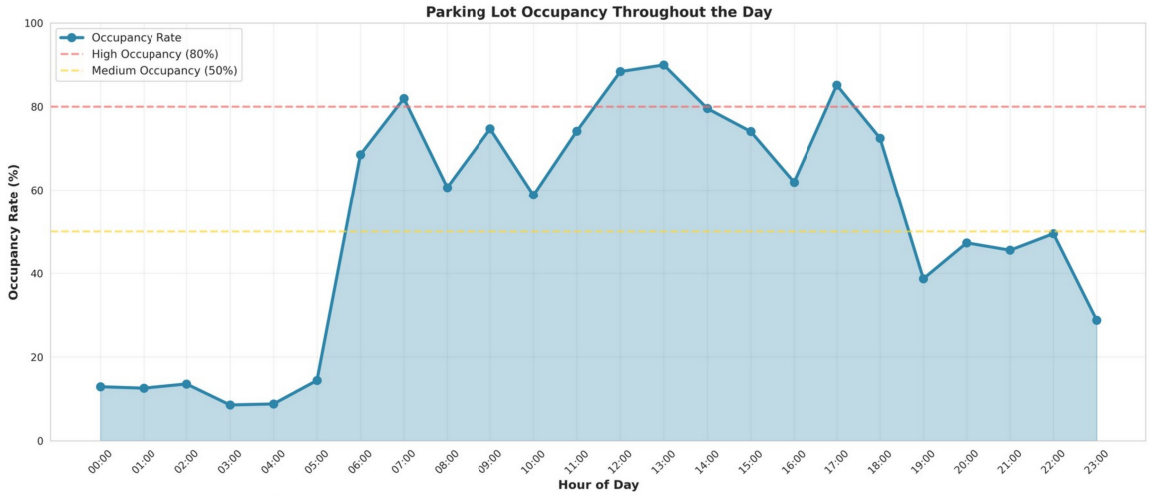


Figure 2: Parking Lot Occupancy Throughout the Day

Figure 2 presents a time-series analysis of occupancy rates over a 24-hour period. The chart clearly highlights peak usage hours, typically showing spikes during morning arrivals (8:00 AM - 9:00 AM) and lunch hours. Threshold lines at 50% and 80% occupancy provide visual benchmarks. When the curve crosses the 80% threshold, the system is designed to trigger 'High Occupancy' alerts to management.

7.3 Performance Metrics Dashboard

The performance metrics dashboard provides a high-level summary of the most critical Key Performance Indicators (KPIs).

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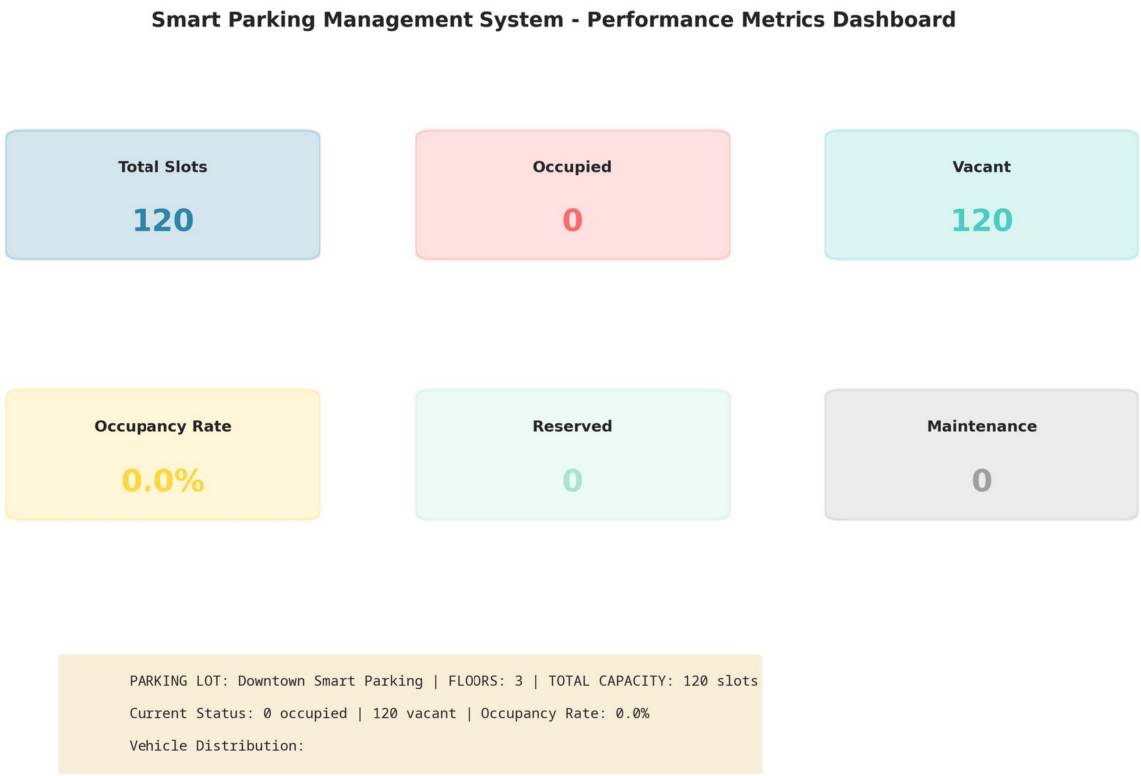


Figure 3: Key Performance Metrics Dashboard

Figure 3 aggregates real-time data into easily readable KPI boxes. It displays the total capacity, current occupied and vacant counts, and the overall occupancy rate percentage. It also tracks slots marked for maintenance or reserved status. The text summary at the bottom provides a quick textual overview of the facility's current operational state, which is ideal for a quick glance by parking attendants.

7.4 Revenue and Vehicle Analysis

Financial tracking is a core requirement for parking operators.

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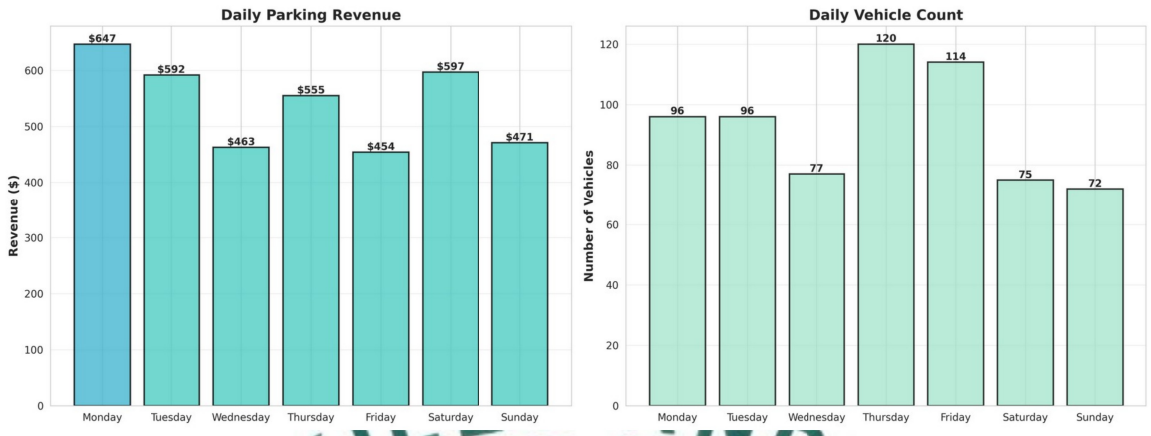


Figure 4: Daily Revenue and Vehicle Count Analysis

Figure 4 correlates the daily financial revenue with the volume of vehicles processed. By analyzing this data, operators can identify which days of the week are most profitable and adjust operational strategies accordingly. The color-coded revenue bars provide an immediate visual indication of performance against daily targets.

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CHAPTER 8

CONCLUSION AND LEARNING EVALUATION

This final chapter addresses Performance Criterion 9 (PC9), summarizing the project outcomes and evaluating the technical skills gained during the internship.

8.1 Project Conclusion

The Real-Time Smart Parking Management System project successfully demonstrated the application of IoT concepts and software engineering to solve a pressing urban challenge. The developed simulation accurately models the complex interactions between vehicles, parking slots, and sensor networks. By integrating advanced analytics and visualization, the system transforms raw occupancy data into actionable intelligence for parking administrators. The solution proves that automated, sensor-driven parking management can significantly enhance space utilization, optimize revenue, and improve the overall driver experience by reducing search times.

8.2 Learning Evaluation (PC9)

Participating in this internship project resulted in substantial technical skill gain and professional development:

- System Architecture Design: I gained practical experience in designing a multi-layered architecture, separating the perception layer (IoT sensors) from the application layer (analytics).
- Advanced Python Programming: My proficiency in Python significantly improved, particularly in using Object-Oriented Programming (OOP) principles, dataclasses, and complex data structures to model real-world entities.
- Data Visualization Skills: I acquired deep knowledge of Matplotlib and Seaborn libraries, learning how to format, style, and generate professional-grade charts and dashboards programmatically.

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- Algorithm Development: Implementing the logic for slot assignment, fee calculation, and sensor anomaly detection enhanced my algorithmic thinking and problem-solving capabilities.
- Software Testing: I learned the importance of rigorous testing, specifically how to simulate edge cases (e.g., lot full scenarios, hardware failures) to ensure system robustness.

Overall, the internship provided a comprehensive, hands-on experience in full-cycle software development, bridging the gap between theoretical academic knowledge and practical industry application.

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CHAPTER 9

REFERENCES

[1] Al-Khatib, A., et al. (2020). "IoT-Based Smart Parking System." IEEE Internet of Things Journal.

[2] Python Software Foundation. (2024). "Python Language Reference, version 3.11." Available at http://www.python.org

[3] Hunter, J. D. (2007). "Matplotlib: A 2D graphics environment." Computing in Science & Engineering, 9(3), 90-95.

[4] Waskom, M. (2021). "Seaborn: statistical data visualization." Journal of Open Source Software, 6(60), 3021.

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